Features and Distractors

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach feature selection with inspectable counterexamples.

Supply the candidate fields, keep exact-rule and learned-model evidence separate and assess the reserved controlled-pair task.

Keep one habit: Ask what stayed fixed, what changed and what the observation actually establishes.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Choose recorded properties that preserve distinctions needed for a stated task

Students investigate representation before **fitting** a model. Keep the paper rule explicit: positive known measurements, fixed orientation, no thickness in the card task, and both boundaries inclusive.

Length and width suffice for that limited rule. A collision between **feature record**s with different correct labels demonstrates lost information; it does not prove that every task needs these same **input**s.

Explain **leakage** through timing and the copied sticker, without requiring model **training** or statistics.

By the end, I can test a distracting **feature** with controlled changes rather than confounding clues.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-1/8, ai-2/9. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Four cards matching the worked table, with dimensions, colour and identifying letters written as text
- Two paper columns headed Before the decision and After inspection, plus removable label stickers
- Blank records headed Length in cm and Width in cm, and the independent display-tray task from W4

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Operational Labels. Give an example and a mistake that the idea helps you avoid.

Look for: Apply a category definition using explicit **evidence** and boundary rules

Reference example, not the only acceptable wording: In order: ready because both known fields pass; revise because 31 exceeds 30; review because the known day passes but the room is unknown; revise because the known day failure has priority over the unknown room.

Equality at 30 is accepted.

Without opening the old lesson, explain one key idea from Debugging Strategies. Give an example and a mistake that the idea helps you avoid.

Look for: Design revealing tests for a function

Reference example, not the only acceptable wording: 4, 0 and 6.

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Look for: Trace the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

Check the card fields against the same explicit policy and version. Agreement alone can preserve a shared mistake; it does not prove that an event exists or has permission.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Operational Labels

Debugging Strategies

Branching Logic

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

All animal books in **training** have green covers. How could you test whether a recommender is using subject or cover colour? Supply two diagnostic test descriptions.

Concept to explain

A **feature** should help the task and be available at decision time.

Term	Meaning
feature	An input attribute or representation used by a model or decision rule.
relevant	Useful to the particular question, target or decision being considered.
distractor	A detail that draws attention but does not help answer the current question.
shortcut	A convenient but unreliable cue used instead of the relationship the task requires.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach features through pairs a system must distinguish

> Keep exact-rule conclusions separate from unknown model behaviour.

State the dimension rule and target

Imagine a paper sorting task for a model delivery station. Each rectangular card has a length, width, colour and card number.

The station labels a card fits if its length is at most 12 centimetres and its width is at most 8 centimetres. Both limits include equality.

Cards stay flat in the marked orientation; turning, bending and thickness are outside this paper task. All measurements in the first examples are known positive whole numbers.

A **feature** is a recorded **input** property, such as width in centimetres. The **target label** is the answer we want to produce: fits or too large under this guide. Decide what the answer means before choosing the **input**s.

A card number can help us find a record, but knowing its number does not tell us its dimensions.

This guide already gives us a direct rule, so we do not need to train a model to solve it. We use the task to investigate the information that either a rule or a **learned model** would need.

Later, when relationships are learned from examples, the same question remains: did we supply the distinctions the task requires?

Cover width, then reveal the distinguishing input

Card A is 10 centimetres long and 7 centimetres wide. Card B is 10 centimetres long and 9 centimetres wide. A fits; B is too large because its width exceeds 8. Compare two ways of recording these cards.

AI ACADEMY / AI-2/9 · 03 · INSPECT THE LESSON MODEL

The same cards under two feature choices

Original card	Choose the inputs	Resulting feature record
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

Follow the labelled picture with the matching explanation. Open the vector diagram

A system receiving only the length cannot use the hidden width to tell these two cards apart. The same **input**, 10, is paired with two different correct answers.

It may guess, return an uncertainty, or use a usual answer, but length alone cannot guarantee the right label for both. A more complicated model cannot reconstruct this missing distinction with certainty from that **input** alone.

A **feature record** needs a consistent order and units. Here we write length in centimetres first, then width in centimetres.

The record 10, 7 means something different if someone silently switches the order or records millimetres. Write the definitions alongside the data.

Model one-factor card comparisons

Card measurements and colour	Label under the guide	What the comparison shows
10 by 7 cm and blue	Fits	Compare with the next card

Card measurements and colour	Label under the guide	What the comparison shows
10 by 9 cm and blue	Too large	Same length and colour but width changes the label
13 by 7 cm and blue	Too large	Same width and colour as the first card but length changes the label
10 by 7 cm and red	Fits	Changing only colour does not change this guide

The first two cards show why width matters. The first and third show why length matters. The first and fourth show that colour has no part in this particular rule.

That is a statement about our task, not a claim that colour is useless for every task. For sorting red and blue cards, colour would be essential.

Suppose every **fitting** card in a small practice set is blue and every oversized card is red. Colour would appear to predict the labels perfectly in that set. A red 10 by 7 card and a blue 13 by 7 card expose the **shortcut**.

Check a proposed **feature** on deliberately different cases before trusting the practice score.

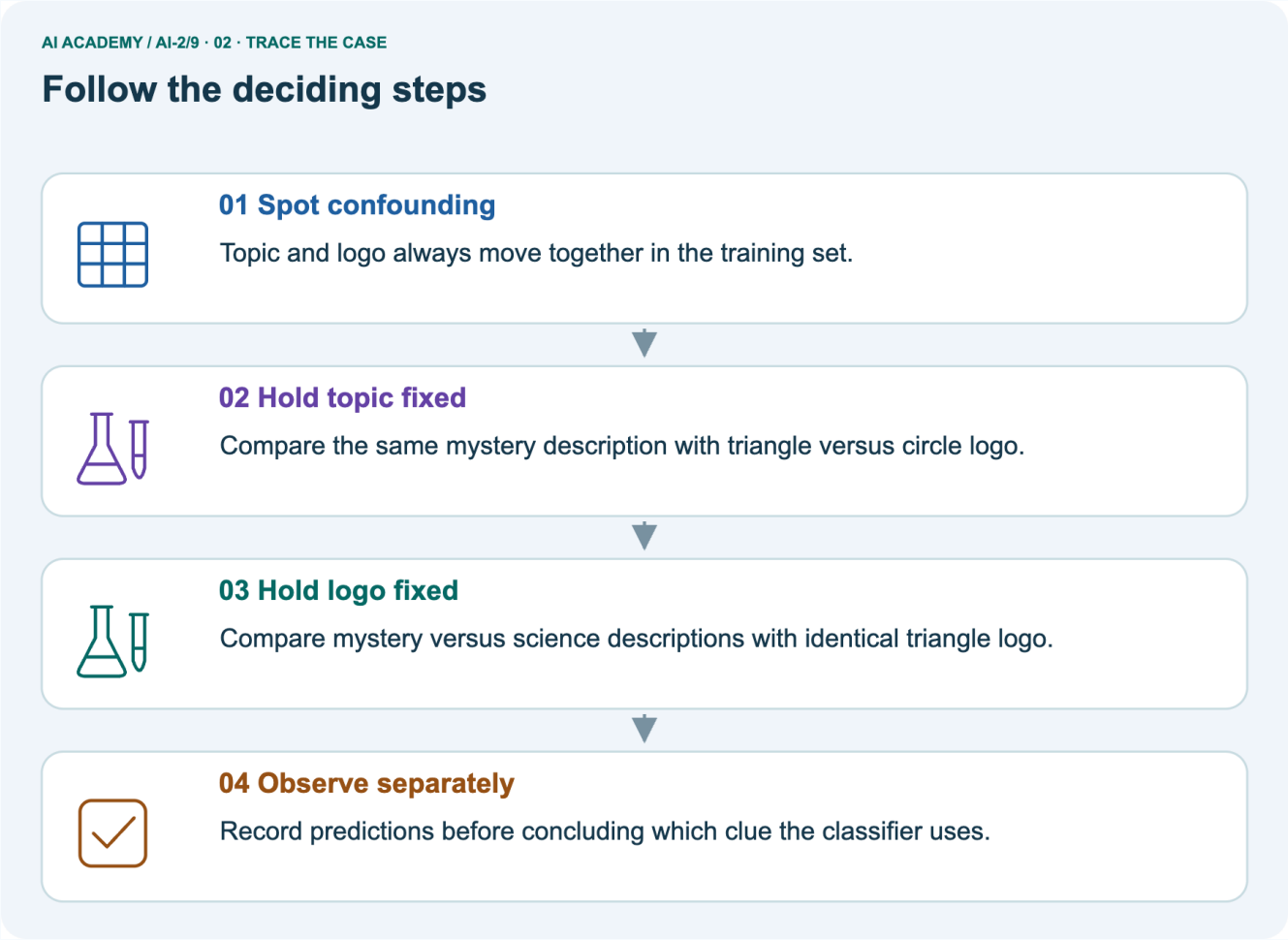
Introduce the separate confounded classifier

A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos.

Training contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

Design controlled pairs before observing results



Follow the labelled picture with the matching explanation. Open the vector diagram

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

Interpret the hypothetical logo-sensitive outcome

Change: The first controlled pair yields mystery for triangle and science for circle.

Prediction: That pair is **evidence** of unwanted logo sensitivity.

Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

Limit the sensitivity claim and use the picture

Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

1. Goal: recommend enjoyable readable books.

2. Useful: genre match, reading level, length preference.

3. **Distractor**: cover color may match past labels by accident.

4. Deployment check: every chosen **feature** must exist before recommending.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Choose useful **evidence** and possible distractions.

Exact explanation to model

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

Explain the added worked example

Key terms

Term	Meaning
Feature	A recorded property supplied as an input to a system
Target label	The answer a supervised learning task asks a model to predict
Feature record	The selected input values for one example, in a defined order
Relevant feature	An input property that helps make distinctions needed for the stated task
Leakage	Information used during development that would not legitimately be available when making the intended prediction

Model the reasoning aloud

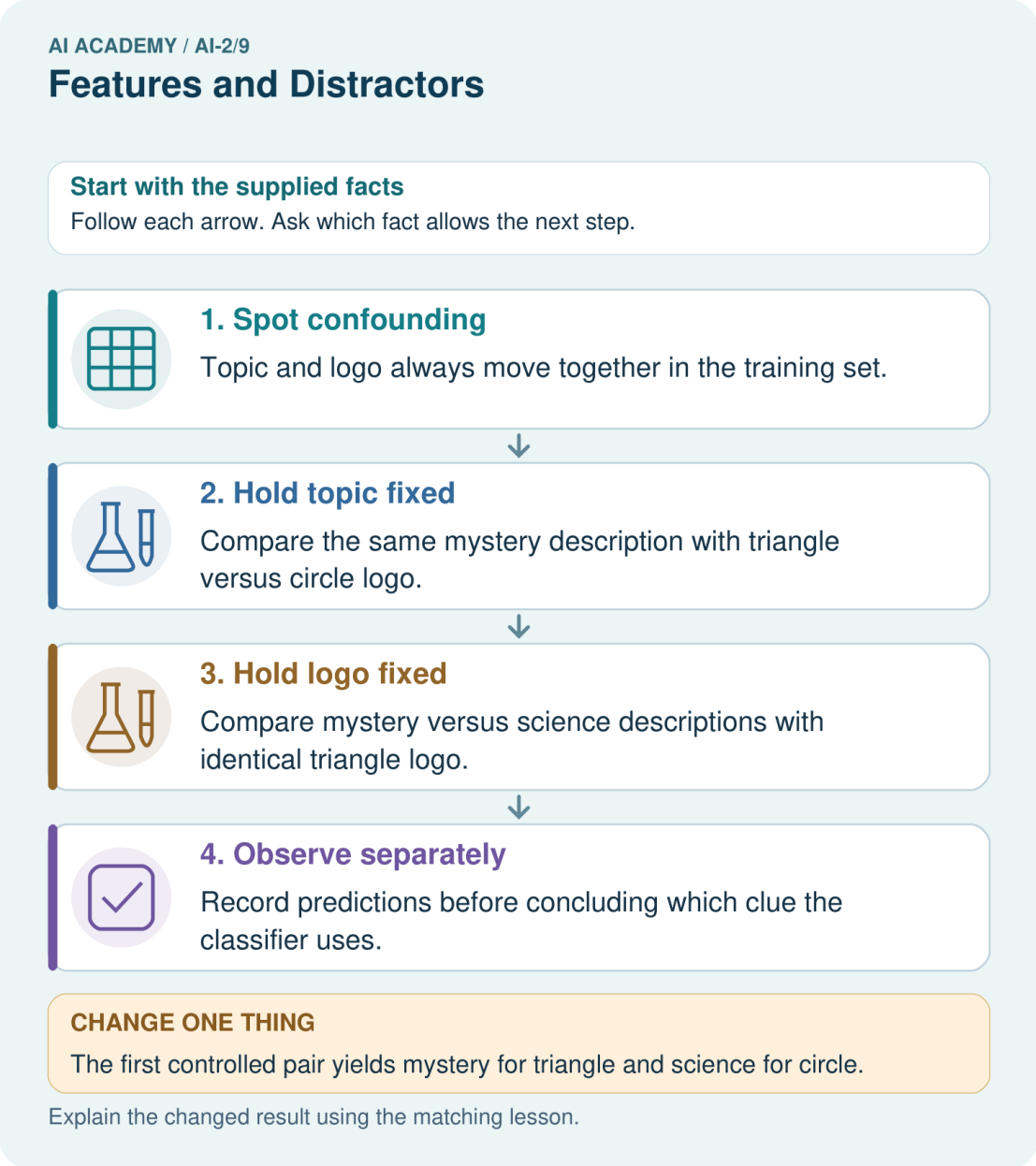
Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Features and Distractors

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Spot confounding	Topic and logo always move together in the training set.
2. Hold topic fixed	Compare the same mystery description with triangle versus circle logo.
3. Hold logo fixed	Compare mystery versus science descriptions with identical triangle logo.
4. Observe separately	Record predictions before concluding which clue the classifier uses.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels.

Explain why a system receiving only width cannot always label both correctly.

Expected reasoning and fading help

Teacher check: One valid pair is length 11 width 6, which fits, and length 13 width 6, which is too large. Both give the width-only **input** 6 but require different labels.

Without length or other permitted distinguishing information, the system cannot guarantee correct labels for both. Accept any valid pair with positive measurements, equal widths at most 8, and lengths on opposite sides of 12.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Cover width and all identifiers for two same-length cards with different labels. Ask which supplied value tells them apart. Guessing or using typical values cannot guarantee correctness for both.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the **relevant** step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

No. In the explicitly hypothetical changed case, one topic-fixed **prediction** changes with logo, supporting sensitivity on that pair. In the base case no new outputs have been supplied, so the designed test alone establishes no observed sensitivity. Neither case proves that the classifier always ignores subject.

Changed-case reasoning: That pair is **evidence** of unwanted logo sensitivity. The subject description stayed mystery; only the ir**relevant** logo changed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark **relevant** details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Target: includes animal characters. Fields: animal tag, cover colour and shelf number.

Write the question to explain: Choose useful **evidence** and possible distractions.

Reveal after **prediction**s: An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

Use the earlier changed question as optional later practice. Reserve Changed independent case for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Length 12 cm is usable before inspection and passes its criterion. The unreadable width remains unknown, so it might be at most 8 or above 8; the available measurements do not establish fits or too large.

The final sticker does not exist before inspection, so it cannot supply an **input** at the time the answer is needed. Using it would leak the later answer.

The exit answer may use red 10 by 7 cm, which fits, or blue 13 by 7 cm, which is too large. Check the dimensions and use new cases that break the practice colour pattern.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start 0 to 7 minutes	Retrieve Week 8: agreement is not proof of a correct label. State the new target and check both inclusive boundaries.
7 to 20 minutes	Show cards A and B. Cover their widths and identifying letters. Ask what identical length-only inputs allow the system to distinguish, then reveal width.
20 to 33 minutes	Trace the four-row comparison table. Have students cite the pair showing each measurement matters and the pair showing colour does not enter this rule.

Stage	Teacher action
33 to 45 minutes	Complete W1 and W2 with partners. Require a constructed pair and reasons; do not accept only a list of feature names.
Session 2 start 0 to 8 minutes	Retrieve the same-input problem using length 11 width 5 and length 11 width 10. Both length-only records are 11; the first fits and the second is too large.
8 to 23 minutes	Use W3 to break the colour shortcut. Place the sticker after the decision line and discuss why its apparent predictive success is misleading.
23 to 38 minutes	Collect W4 independently. Look for height and thickness, a valid equal-height pair, boundary reasoning and the timing explanation.
38 to 45 minutes	Use W5 and the exit response to revisit task-dependent relevance. Preview algorithms that update a recorded state one step at a time.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

For the eight-field lab select length_cm and width_cm. Colour, identifier, thickness and record time are unnecessary for this stated rule; learner name is unnecessary personal information; the later sticker leaks the target. Do not demand a third selected feature. The controlled dimension pairs have fits/too-large labels; both colour copies of 10×7 fit. The separate topic-logo base case supplies no model outputs. Interpret only the explicitly

hypothetical changed-case results as evidence of logo sensitivity on that pair, not universal dependence. Reserve the space/animal controlled-comparison case for the first independent attempt.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Audit the eight supplied candidate **feature**s for relevance, availability, privacy and **shortcut** risk. Defend the smallest sufficient set for the written card rule and reject at least two tempting alternatives; do not add a third **feature** merely to meet a quota.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

A stronger model can always recover an omitted measurement: Cover width and all identifiers for two same-length cards with different labels. Ask which supplied value tells them apart.

Guessing or using typical values cannot guarantee correctness for both.

A **feature** that matches practice labels must be useful: Use a red **fitting** card or a blue oversized card. Ask whether the **feature** preserves the stated distinction when the practice pattern changes.

The final label is the best **input feature**: Place the sticker after inspection on the timeline. It is a target for checking or teaching, not an available **input** for a **prediction** before inspection.

Useful **feature**s stay useful for every task: Change the target to red or blue bins. Identical dimensions can now require different answers, so colour matters.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Changed independent case for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: **Training** covers show space books only on yellow backgrounds and animal books only on blue backgrounds. Design two controlled comparisons to test whether a classifier uses the intended subject.

Compare different subjects on the same background, then the same subject on different backgrounds, holding other **feature**s fixed. Record actual **prediction**s before claiming a **shortcut**.

The **training** pattern alone makes the **shortcut** plausible, not established.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the **relevant** skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Separate inputs from the target

For the fixed-orientation card task, fits requires length at most 12 cm and width at most 8 cm, including equality.

Choose the measurements needed as **input**s from length, width, colour, card number and the final inspection sticker. State the **target label** and write the **feature record** for a 12 cm by 8 cm red card using a defined order.

Use length and width measured before the decision. State the order as length in cm, then width in cm; the record is 12, 8.

The target is fits or too large under the guide, and this card is fits because both equalities are included.

Colour and card number are not needed for this rule; the final inspection sticker copies the answer and is unavailable before inspection.

W2 Prove that one measurement is insufficient

Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels. Explain why a system receiving only width cannot always label both correctly.

One valid pair is length 11 width 6, which fits, and length 13 width 6, which is too large. Both give the width-only **input** 6 but require different labels.

Without length or other permitted distinguishing information, the system cannot guarantee correct labels for both. Accept any valid pair with positive measurements, equal widths at most 8, and lengths on opposite sides of 12.

W3 Challenge an apparent colour shortcut

A practice set contains three blue cards that fit and three red cards that are too large. A helper uses blue means fits and red means too large. Create a new card that this helper mislabels.

Explain why the perfect practice score did not establish that colour was a reliable **feature** for this task.

A red 10 by 7 cm card fits but the helper says too large; alternatively a blue 13 by 7 cm card is too large but the helper says fits. The practice set paired colour with size and did not test cases that separate them.

The stated guide depends on measurements, so compare **prediction**s with that guide on cases where colour and size no longer follow the **shortcut**.

W5 Notice when the task changes

The delivery station now sorts cards into red and blue bins instead of checking dimensions. Which previously unneeded **feature** becomes useful? Explain why length and width alone may now be insufficient.

Give two cards that support your explanation.

Colour becomes useful because the new target is red or blue. A red 10 by 7 card and a blue 10 by 7 card have identical length-width records but require different bins.

Feature relevance depends on the task; reusing the previous **feature** set without reviewing the target can discard the needed distinction.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

An accurate animal tag directly addresses the target. Colour and shelf can produce accidental **shortcut**s. Test on new books.

Additional transfer practice question

Predict whether a book is short using page count and cover colour. Which **feature** directly supports the defined label? Give a counterexample to colour.

Reasoned practice answer

Page count directly supports the definition. A blue book can have 40 or 200 pages, so colour alone does not determine length.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **feature** that appeared often in **training** must be causal.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Use a non-green animal book and a green non-animal book, with trusted subject labels. Subject-based **prediction**s should follow animal content rather than cover colour.

Retain varied held-out books; two examples diagnose a **shortcut** without proving general reliability.

Explanations for the additional workbook examples

W1 Separate inputs from the target

W2 Prove that one measurement is insufficient

W3 Challenge an apparent colour shortcut

W4 Design a record for a new task

A fictional display tray accepts a tile if its height is at most 9 cm and its thickness is at most 2 cm, including equality. Other dimensions are fixed. Before checking, you can record height, thickness and pattern name.

A pass sticker is added afterward. Choose **input**s with units. Construct two tiles with the same height but different labels.

Then label a 9 cm high, 2 cm thick striped tile and explain why the sticker is unsuitable as an **input** before checking.

Record height in cm and thickness in cm. A valid pair is height 8 thickness 1, which passes, and height 8 thickness 3, which fails: equal height hides the required thickness distinction.

The 9 cm high, 2 cm thick tile passes both inclusive boundaries. Pattern name does not affect the stated rule. The pass sticker is created after checking and copies the target; using it to predict before checking is **leakage**.

W5 Notice when the task changes

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Use explainable traits that connect to the reader's stated preferences.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Let students physically cover unselected columns and compare two records side by side. Provide the sentence frame Same **input** ___ but labels ___ and ___ because ___. Use written colour names so no task depends on colour vision.

Extension: ask learners to prove the width-only **input** fails using a second valid pair, or explain how permitting card rotation would change the decision procedure even when the recorded dimensions stay the same.

For the selected W4, require two controlled comparisons: hold the subject description fixed while changing only the yellow/blue background, then hold background fixed while comparing the two supplied subjects, with other presentation features controlled. Record expected subject labels separately from actual predictions, which have not been supplied. The old training pattern suggests a possible shortcut but does not establish one. After feedback use a different subject/style pairing for a fresh attempt. The display-tray case is optional transfer practice with a separate key.

Plan the delayed check

Suggested return points in this level: ai-2/10, ai-2/12, ai-2/15. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 10: Human Classifier Simulation. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 10: Human Classifier Simulation. Carry forward explicit input features, the target definition and a record of predictions versus supplied evidence; keep the rule or simulation inspectable.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Choose recorded properties that preserve distinctions needed for a stated task

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Invent two cards with the same width but different correct labels under the card guide. Give both lengths and widths and justify the labels.

Explain why a system receiving only width cannot always label both correctly.

Reasoned practice answer: One valid pair is length 11 width 6, which fits, and length 13 width 6, which is too large. Both give the width-only **input** 6 but require different labels.

Without length or other permitted distinguishing information, the system cannot guarantee correct labels for both. Accept any valid pair with positive measurements, equal widths at most 8, and lengths on opposite sides of 12.

Check every learner before discussion: Use the worked case for **Feature**s and useful distinctions. Proposed explanation: “A stronger model can always recover an omitted measurement”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Cover width and all identifiers for two same-length cards with different labels. Ask which supplied value tells them apart. Guessing or using typical values cannot guarantee correctness for both.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The first controlled pair yields mystery for triangle and science for circle. Require a **prediction** and a reason before comparing. Expected result: That pair is **evidence** of unwanted logo sensitivity.

Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Make a transparent first version. Keep a trace of how one **input** becomes an **output**; explain the rule or learned component.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ Earlier wording, alternatives and picture transcripts

This week has no code block. Use this routine when a later lesson includes an executable lab.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

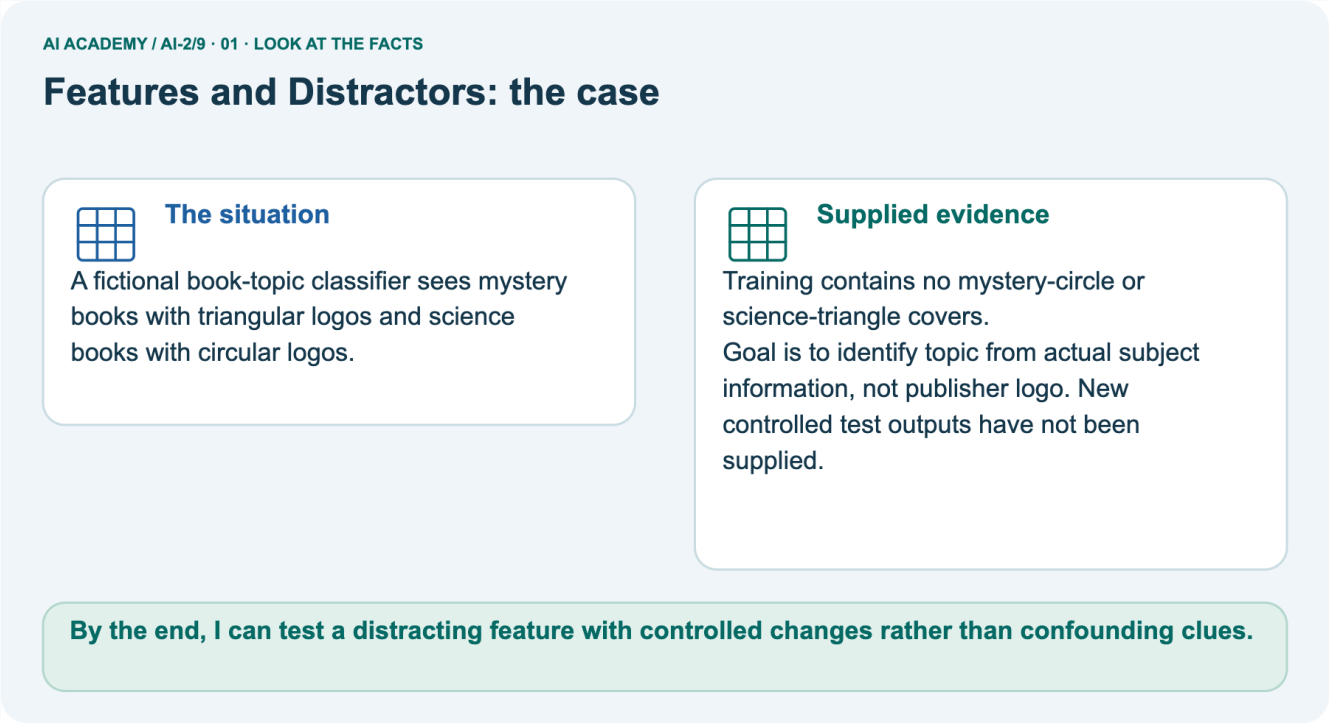
▼ Reference: teach with the six pictures

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Features and Distractors: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

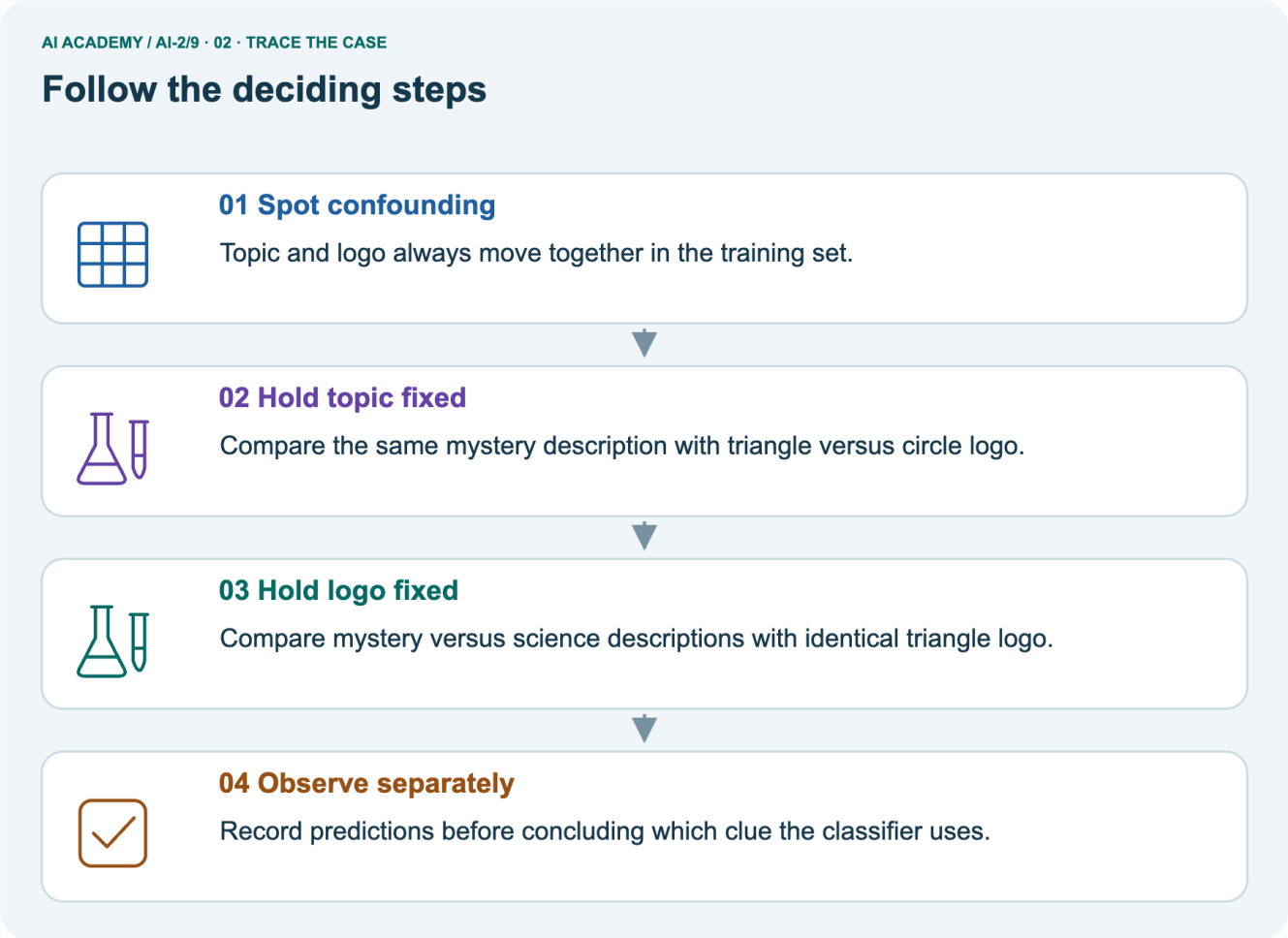
A fictional book-topic classifier sees mystery books with triangular logos and science books with circular logos. Supplied facts: **Training** contains no mystery-circle or science-triangle covers.

Goal is to identify topic from actual subject information, not publisher logo. New controlled test **output**s have not been supplied.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



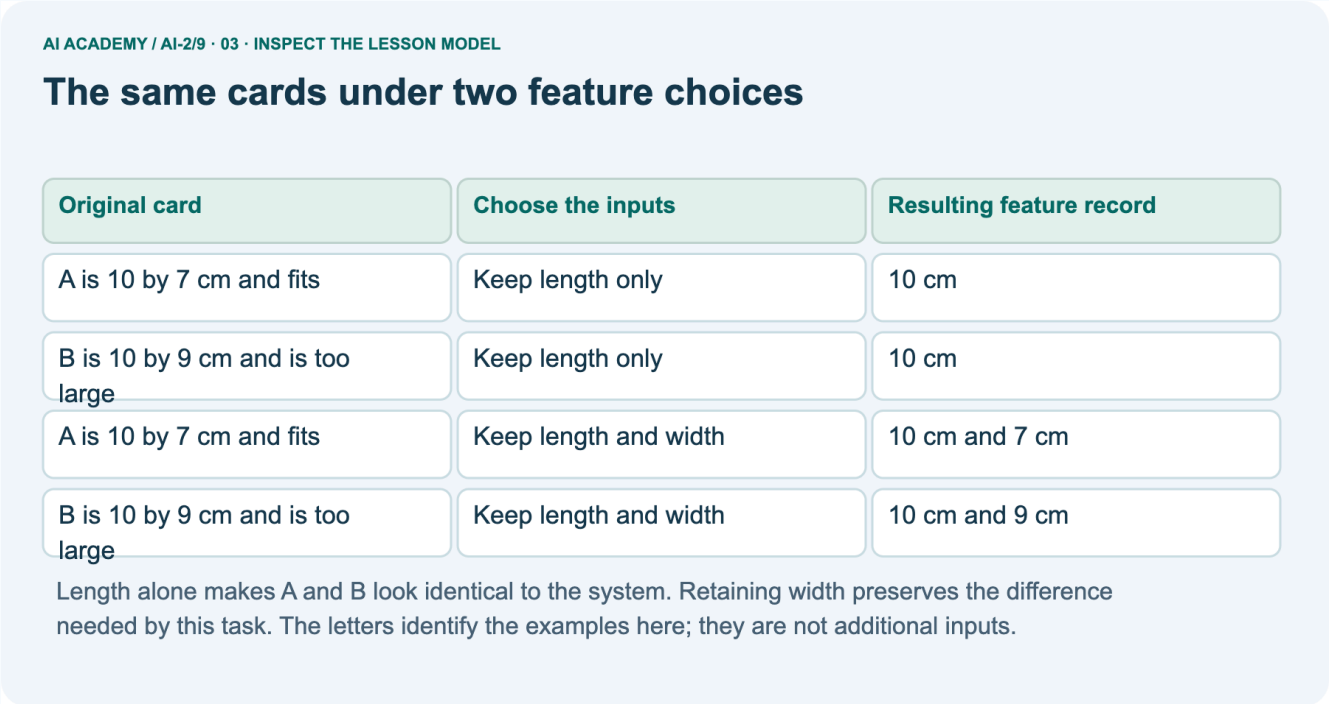
Follow the labelled picture with the matching explanation. Open the vector diagram

Spot confounding: Topic and logo always move together in the **training** set. → Hold topic fixed: Compare the same mystery description with triangle versus circle logo. → Hold logo fixed: Compare mystery versus science descriptions with identical triangle logo. → Observe separately: Record **prediction**s before concluding which clue the classifier uses.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. The same cards under two feature choices



AI ACADEMY / AI-2/9 · 03 · INSPECT THE LESSON MODEL

The same cards under two feature choices

Original card	Choose the inputs	Resulting feature record
A is 10 by 7 cm and fits	Keep length only	10 cm
B is 10 by 9 cm and is too large	Keep length only	10 cm
A is 10 by 7 cm and fits	Keep length and width	10 cm and 7 cm
B is 10 by 9 cm and is too large	Keep length and width	10 cm and 9 cm

Length alone makes A and B look identical to the system. Retaining width preserves the difference needed by this task. The letters identify the examples here; they are not additional inputs.

Follow the labelled picture with the matching explanation. Open the vector diagram

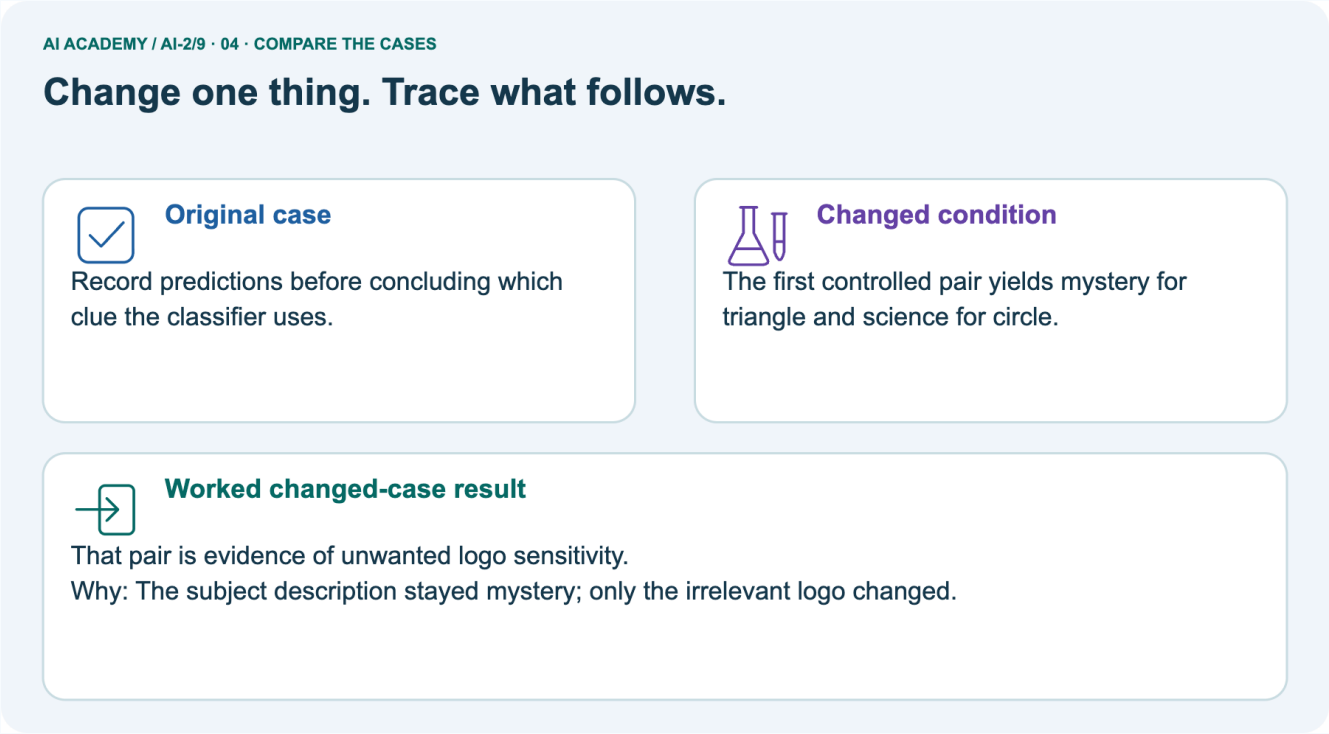
The same cards under two **feature** choices. A is 10 by 7 cm and fits | Keep length only | 10 cm. B is 10 by 9 cm and is too large | Keep length only | 10 cm. A is 10 by 7 cm and fits | Keep length and width | 10 cm and 7 cm.

B is 10 by 9 cm and is too large | Keep length and width | 10 cm and 9 cm

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Record **prediction**s before concluding which clue the classifier uses. Changed condition: The first controlled pair yields mystery for triangle and science for circle.

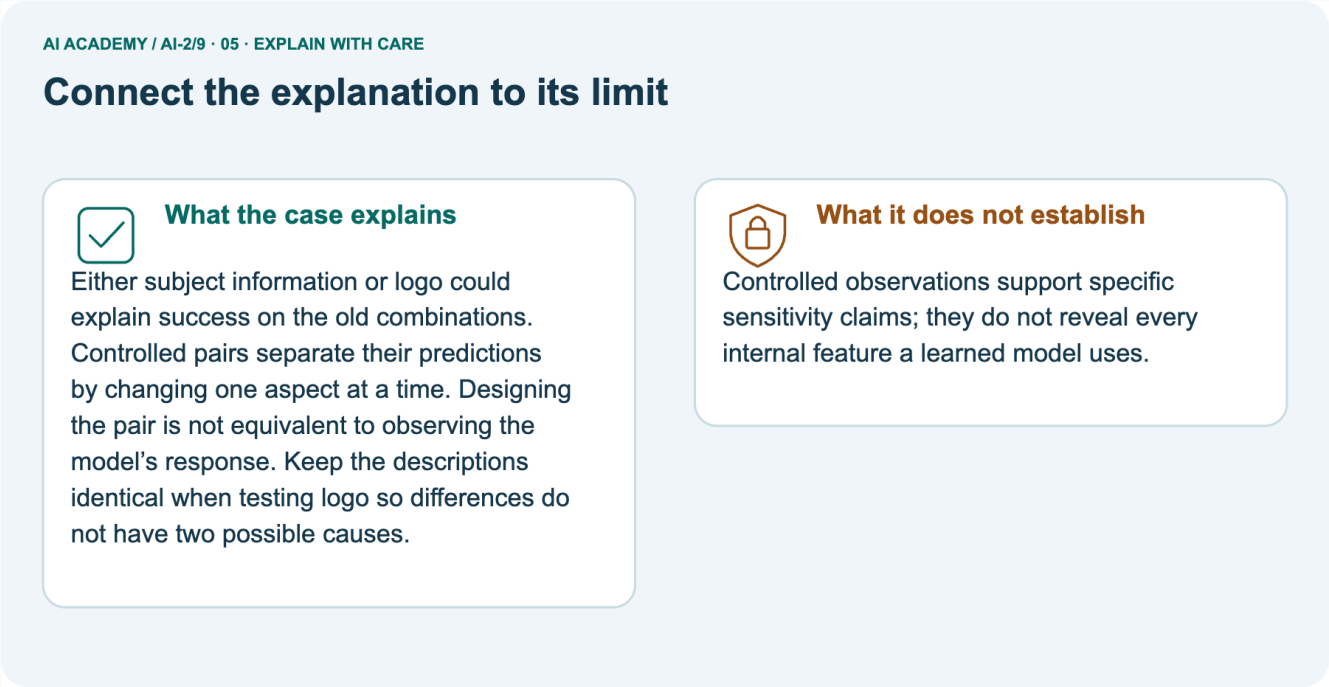
Expected result: That pair is **evidence** of unwanted logo sensitivity. Reason: The subject description stayed mystery; only the ir**relevant** logo changed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The subject description stayed mystery; only the ir**relevant** logo changed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Either subject information or logo could explain success on the old combinations. Controlled pairs separate their **prediction**s by changing one aspect at a time.

Designing the pair is not equivalent to observing the model's response. Keep the descriptions identical when testing logo so differences do not have two possible causes.

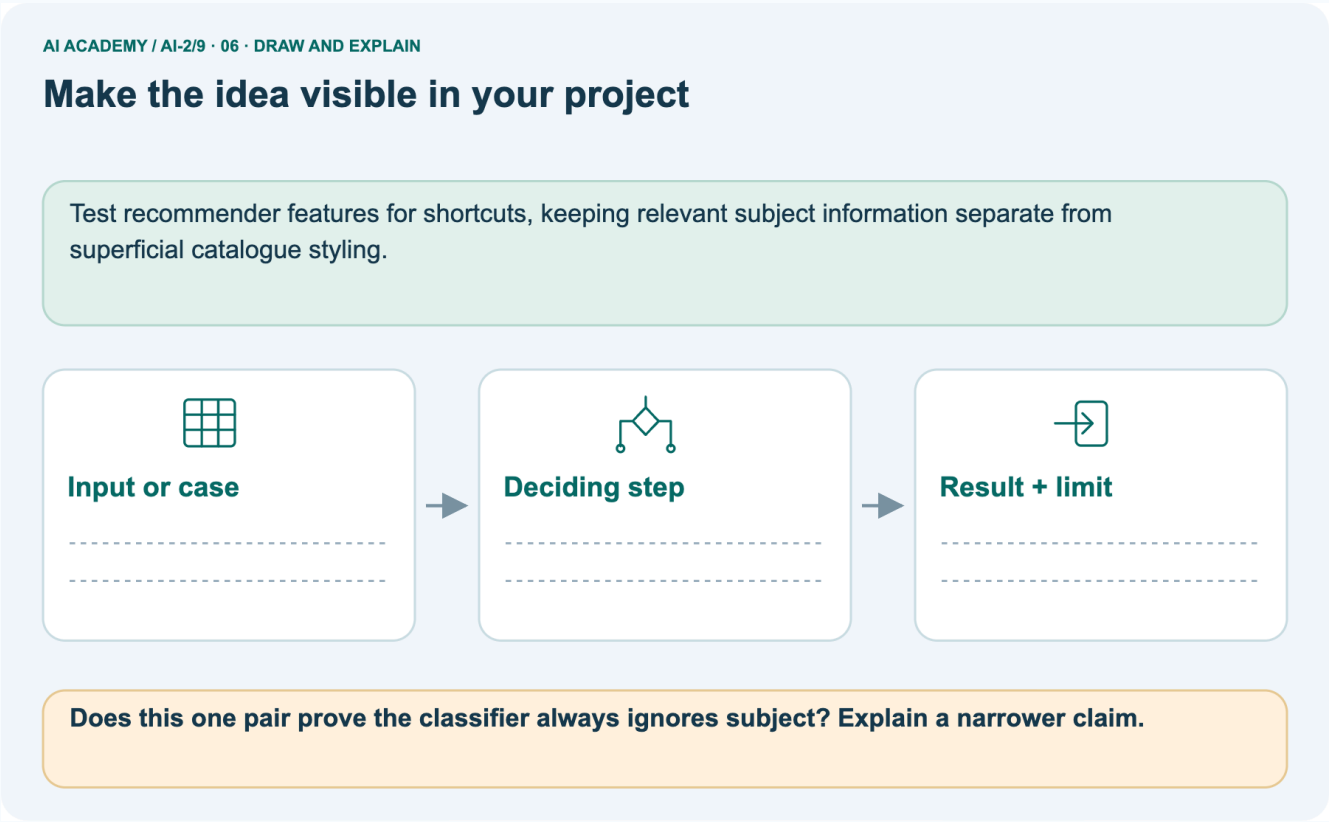
Limit: Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Controlled **observation**s support specific sensitivity claims; they do not reveal every internal **feature** a **learned model** uses.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Test recommender **feature**s for **shortcut**s, keeping **relevant** subject information separate from superficial catalogue styling. Draw your **input**, deciding step, result and limit.

Supported question: Does this one pair prove the classifier always ignores subject? Explain a narrower claim.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see. Supported practice key: No. In the explicitly hypothetical changed case, one topic-fixed **prediction** changes with logo, supporting sensitivity on that pair. In the base case no new outputs have been supplied, so the designed test alone establishes no observed sensitivity. Neither case proves that the classifier always ignores subject.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Features and Distractors: the case

02. Follow the deciding steps

03. The same cards under two feature choices

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.